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Miyachi et al.

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(54) **SWITCH CIRCUIT**

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B60R 16/027	(2006.01)
H03K 17/12	(2006.01)
B60N 2/02	(2006.01)

(52) **U.S. Cl.**

CPC ***H03K 17/122*** (2013.01); ***B60R 16/027*** (2013.01); *B60N 2/0224* (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,456,861 A *	6/1984	Ratzel	B60N 2/0248	318/568.1
4,967,178 A *	10/1990	Saito	B60N 2/0228	340/460
7,239,096 B2 *	7/2007	Hancock	B60N 2/0252	318/59

FOREIGN PATENT DOCUMENTS

JP	50-27058	3/1975

* cited by examiner

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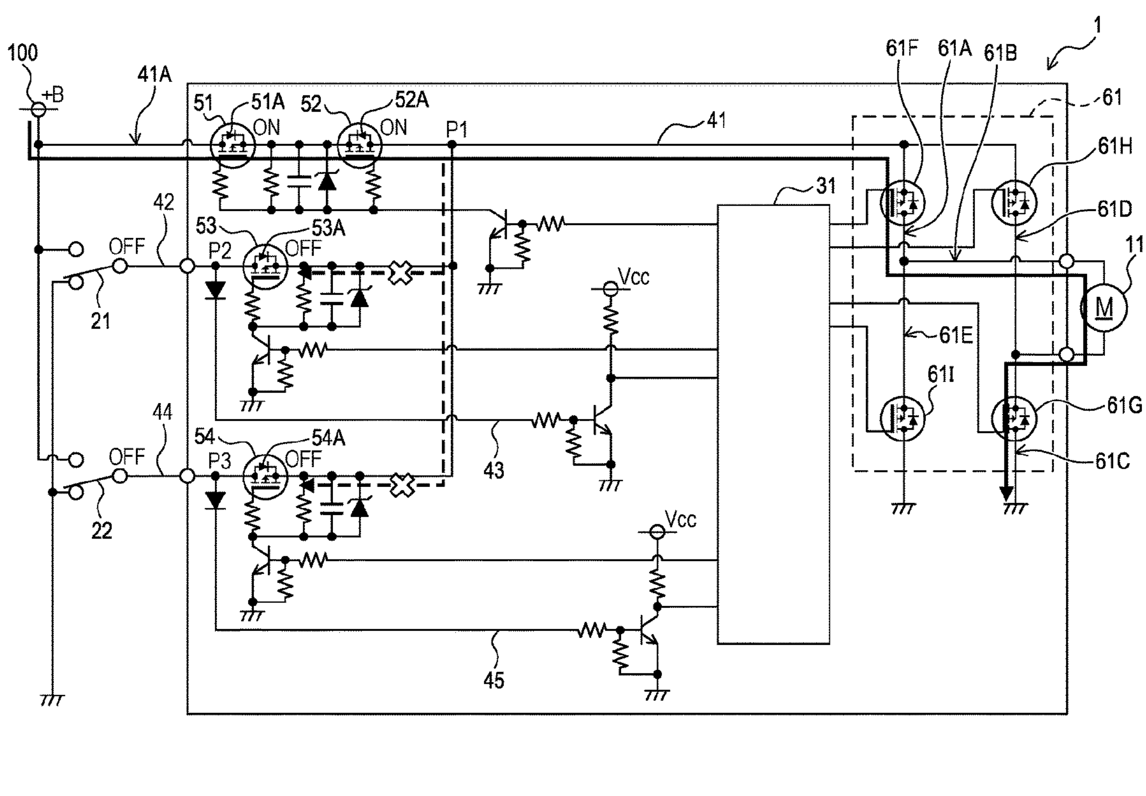
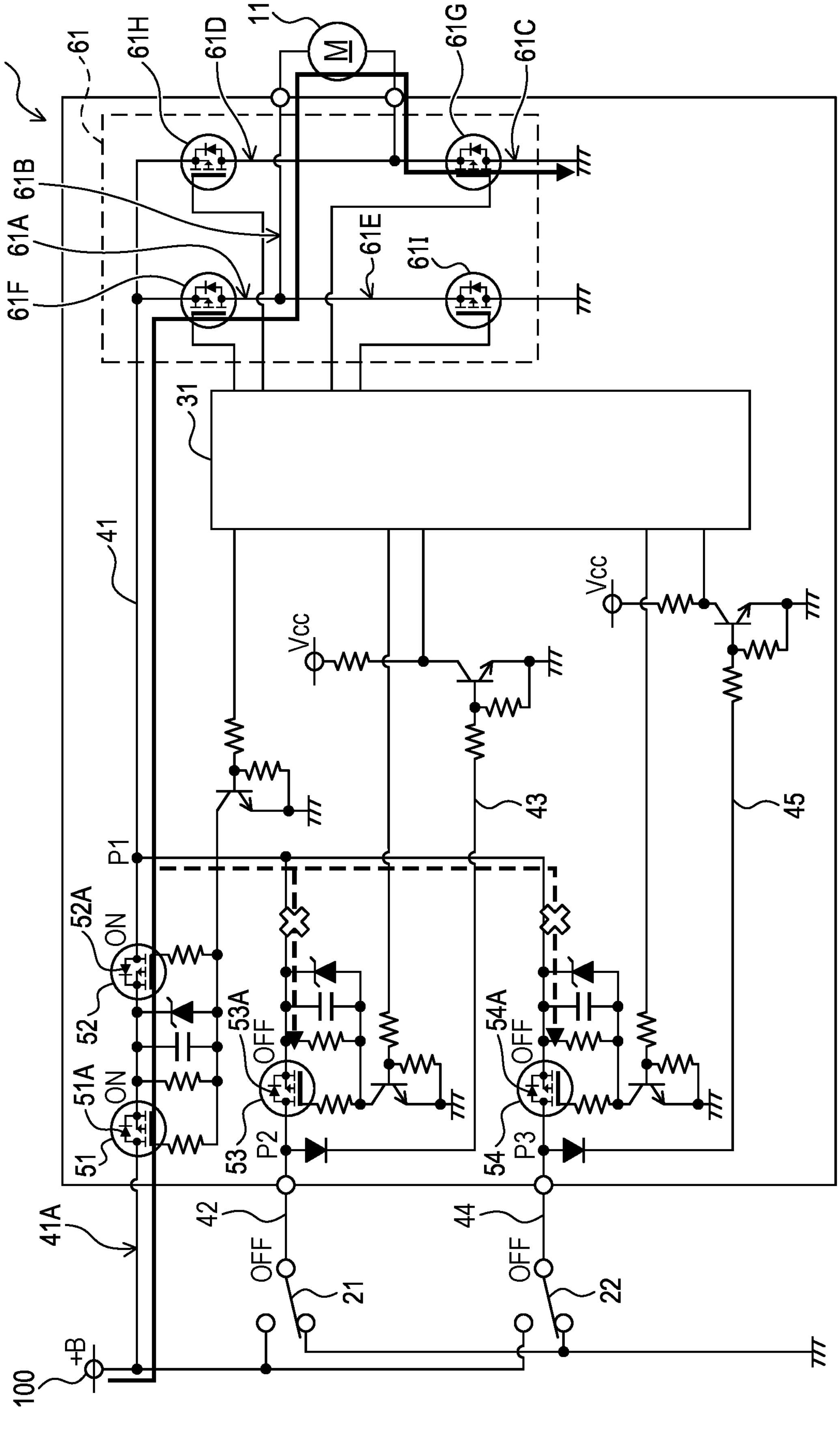
(57) **ABSTRACT**

A switch circuit includes a switch element, a control circuit, a supply line, a switch line that connects the switch element in parallel to a parallel part of the supply line, an input line that allows input of a current from the switch line to the control circuit, a first FET and a second FET provided in the parallel part, and a third FET provided in the switch line in parallel to the input line. The second FET includes a body diode having a forward direction opposite to a forward direction of a body diode of the first FET. The third FET includes a body diode having a forward direction coincident with a direction in which a current is supplied to the load via the switch element.

6 Claims, 5 Drawing Sheets

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43
45
P1
52A
ON
52
53A
OFF
54A
OFF
51A
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51
53
P2
54
P3
41A
42
44
OFF
21
OFF
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+B
100

FIG. 1

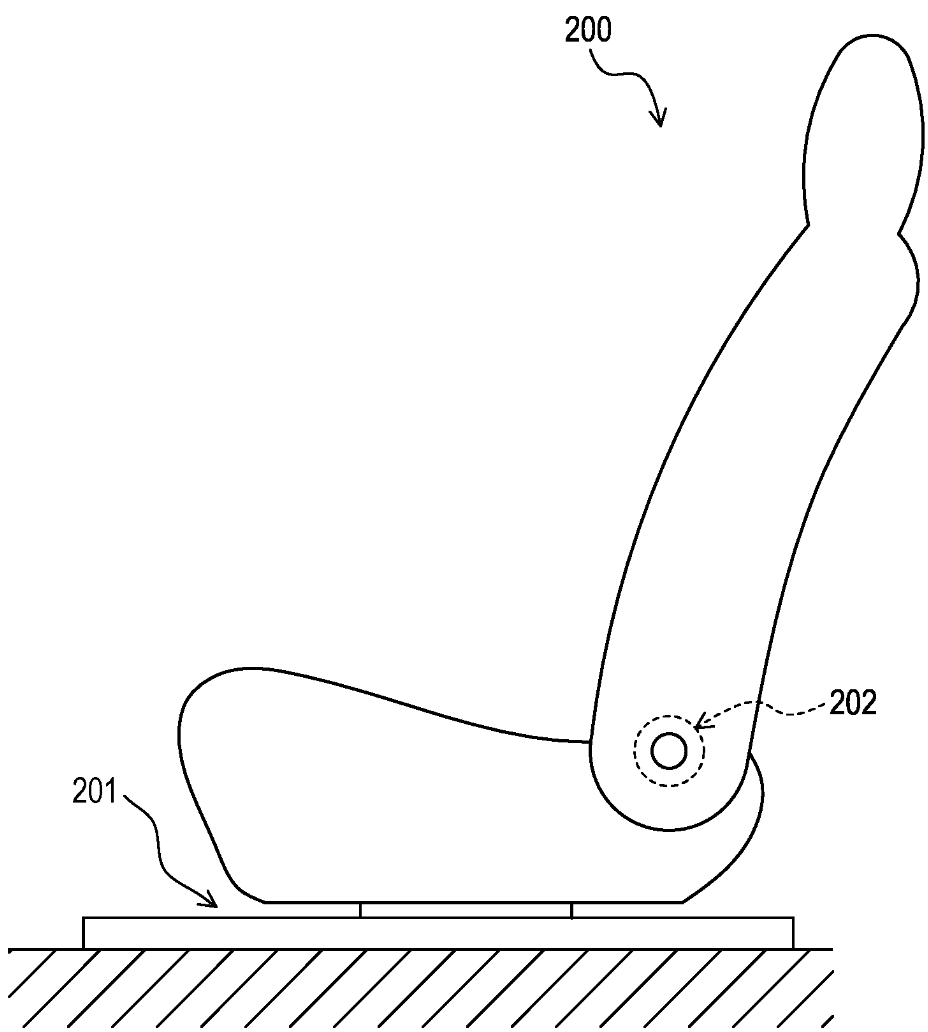

FIG. 2

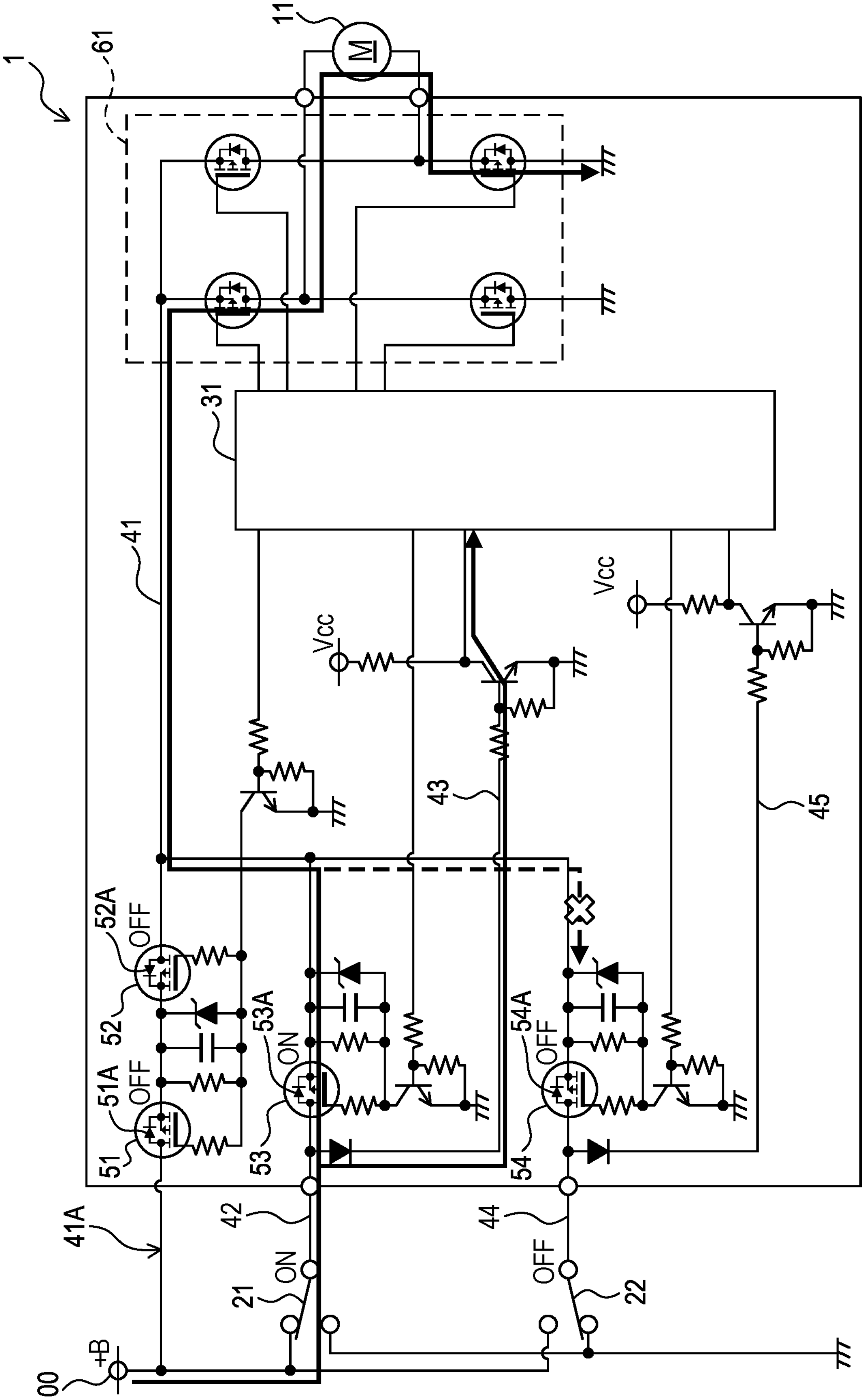
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FIG. 3

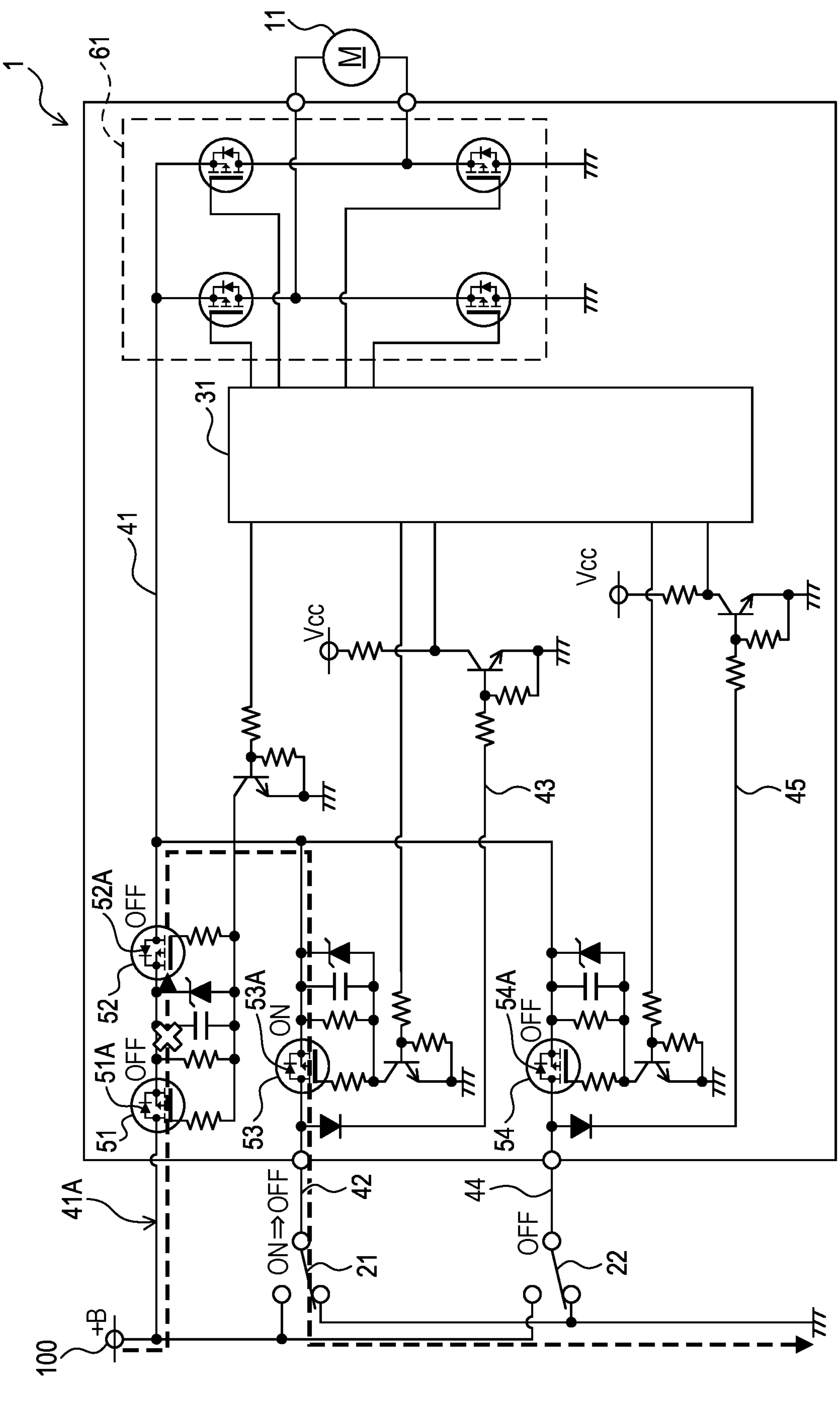
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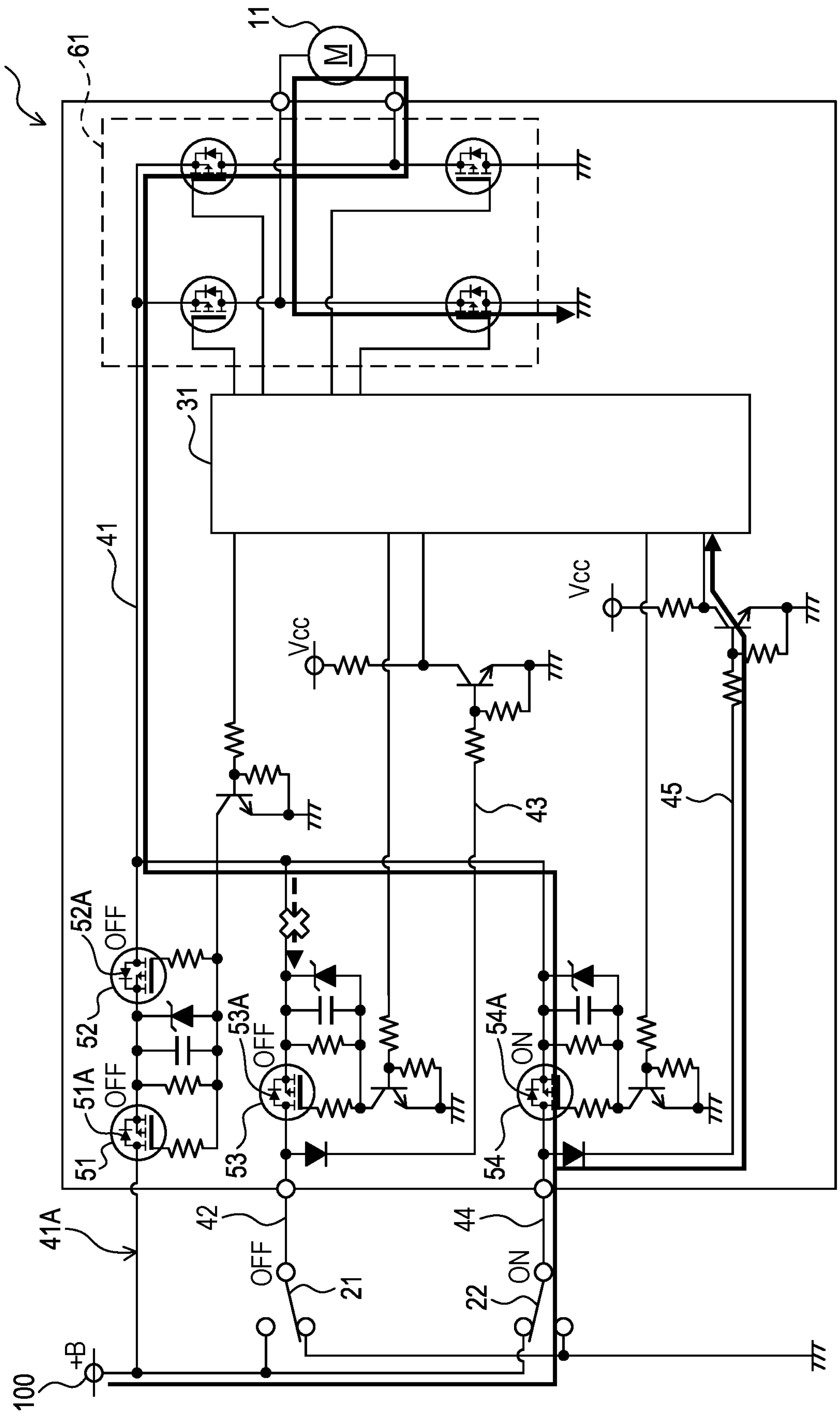
FIG. 4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52
51A
OFF
51
53A
OFF
53
54A
ON
54
OFF
21
ON
22
+B
100

FIG. 5

SWITCH CIRCUIT

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application claims priority based on Japanese Patent Application No. 2023-001797 filed to the Japan Patent Office on Jan. 10, 2023, and the entire content of Japanese Patent Application No. 2023-001797 is incorporated herein by reference.

BACKGROUND

The present disclosure relates to a switch circuit.

If a minute current flows through a switch element in a switch circuit for supplying a current to a load such as a motor, the minute current causes removal of an insulating film, which has been used for protection of the switch element (see, for example, Japanese Unexamined Patent Application Publication No. S50-27058).

SUMMARY

In a conventional switch circuit, when a large current flows through a switch element, loss of the circuit increases. Thus, when a current is supplied to a load via the switch element, there may be a limit to magnitude of the current.

One aspect of the present disclosure has an object to provide a switch circuit that can allow a large current to flow through a switch element.

One aspect of the present disclosure is a switch circuit comprising a first switch element, a control circuit, a supply line configured to allow supply of a current from a power source to a load, a first switch line that connects the first switch element in parallel to a parallel part, which is a part of the supply line, a first input line configured to allow input of a current from the first switch line to the control circuit, a first FET and a second FET provided in the parallel part, and a third FET provided in the first switch line in parallel to the first input line.

The first FET comprises a first body diode having a forward direction coincident with a direction in which a current is supplied to the load. The second FET comprises a second body diode having a forward direction opposite to the forward direction of the first body diode. The third FET comprises a third body diode having a forward direction coincident with a direction in which a current is supplied to the load via the first switch element.

The control circuit is configured to switch on the first FET and the second FET and switch off the third FET when the control circuit receives no current input from the first input line, and to switch off the first FET and the second FET and switch on the third FET when the control circuit receives a current input from the first input line.

In such a configuration, when the first switch element is closed, the first FET and the second FET turn off, so that the current does not flow through the parallel part but is supplied from the first switch line provided with the first switch element to the load. This leads to interruption of the current from the power source in the parallel part, so that a large current can be caused to flow through the first switch element. Further, since a current is inhibited from flowing from the parallel part into the first input line, the control circuit can detect opening and closing of the first switch element.

When the first switch element is open, the third FET turns off, so that the current does not flow through the first switch line but is supplied from the parallel part to the load. Thus, a current is inhibited from flowing from the parallel part into the first input line, and the control circuit can detect opening and closing of the first switch element.

One aspect of the present disclosure may further comprise a second switch element, a second switch line that connects the second switch element in parallel to the parallel part, a second input line configured to allow input of a current from the second switch line to the control circuit, and a fourth FET provided in the second switch line in parallel to the second input line. The fourth FET comprises a fourth body diode having a forward direction coincident with a direction in which a current is supplied to the load via the second switch element.

The control circuit may be configured to switch on the first FET and the second FET and switch off the third FET and the fourth FET when the control circuit receives no current input from either the first input line or the second input line, and to switch off the first FET, the second FET and the third FET and switch on the fourth FET when the control circuit receives a current input from the second input line.

In such a configuration, it is possible to cause a large current to flow through each of the first switch element and the second switch element. Further, the control circuit can detect opening and closing of each of the first switch element and the second switch element.

One aspect of the present disclosure may further comprise a current switching circuit provided in the supply line between a connection point, at which the first switch line and the second switch line are connected, and the load. The second switch element may be configured not to be closed concurrently with the first switch element. The control circuit may be configured to control the current switching circuit to cause a current to flow through the load in a first direction when the control circuit receives the current input from the first input line, and to control the current switching circuit to cause a current to flow through the load in a second direction opposite to the first direction when the control circuit receives the current input from the second input line.

In such a configuration, a selection operation between the first switch element and the second switch element enables a change in the direction of the current to be supplied to the load.

One aspect of the present disclosure may be arranged in a vehicle seat. In such a configuration, it is possible to supply a current to the load that transforms the vehicle seat while removing an insulating film of the first switch element.

BRIEF DESCRIPTION OF THE DRAWINGS

An embodiment of the present disclosure will be described hereinafter by way of examples with reference to the accompanying drawings, in which:

FIG. **1** is a schematic circuit diagram of a switch circuit in an embodiment;

FIG. **2** is a schematic perspective view of a vehicle seat in an embodiment;

FIG. **3** is a schematic circuit diagram showing a state in which a first switch element is closed in the switch circuit of FIG. **1**;

FIG. **4** is a schematic circuit diagram showing a state immediately after the first switch element is opened in the switch circuit of FIG. **1**; and

FIG. **5** is a schematic circuit diagram showing a state in which a second switch element is closed in the switch circuit of FIG. **1**.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

Hereinafter, embodiments applied with the present disclosure will be described with reference to the drawings.

1. First Embodiment

1-1. Configuration

A switch circuit **1** shown in FIG. **1** comprises a load **11**, a first switch element **21**, a second switch element **22**, a control circuit **31**, a supply line **41**, a first switch line **42**, a first input line **43**, a second switch line **44**, a second input line **45**, a first FET **51**, a second FET **52**, a third FET **53**, a fourth FET **54**, and a current switching circuit **61**.

The switch circuit **1** is arranged in a vehicle seat **200** shown in FIG. **2**. The vehicle seat **200** is installed in a vehicle such as an automobile, a railroad vehicle, a ship/boat, or an aircraft. The switch circuit **1** is used for applications, for example, for driving a sliding device **201** or a recliner **202** of the vehicle seat **200**.

<Load>

The load **11** shown in FIG. **1** is, for example, a motor to shift the sliding device **201** or the recliner **202**. The load **11** is supplied with a current from a power source **100** included in the vehicle installed with the vehicle seat **200**.

<First Switch Element>

The first switch element **21** is configured to cause a current to flow through the load **11** in a first direction in a closed state of the first switch element **21**.

The first switch element **21** includes a contact to be connected to the power source **100** in the closed state of the first switch element **21**, and to be earthed in an open state of the first switch element **21**. The first switch element **21** is opened and closed, for example, in conjunction with a physical switch for sliding the vehicle seat **200** frontward.

<Second Switch Element>

The second switch element **22** is configured to cause a current to flow through the load **11** in a second direction opposite to the first direction, in a closed state of the second switch element **22**.

The second switch element **22** includes a contact to be connected to the power source **100** in the closed state of the second switch element **22**, and to be earthed in an open state of the second switch element **22**. The second switch element **22** is opened and closed, for example, in conjunction with the physical switch for sliding the vehicle seat **200** rearward.

The second switch element **22** is configured not to be closed concurrently with the first switch element **21** by a physical structure (e.g., a lock mechanism). Thus, the second switch element **22** can be closed only when the first switch element **21** is open.

<Control Circuit>

The control circuit **31** is an interface circuit configured to individually switch on and off a plurality of Field Effect Transistors (FETs) included in the switch circuit **1**, in response to receiving an input of a signal current. The control circuit **31** is connected to a gate of each FET.

The control circuit **31** is connected with the first input line **43**, the second input line **45**, and a plurality of output lines connected to the gates of the FETs respectively. As the control circuit **31**, for example, a known microcomputer may be used. Specific control of the control circuit **31** will be described below.

<Supply Line>

The supply line **41** allows supply of the current from the power source **100** to the load **11**. Specifically, the supply line **41** connects the power source **100** and the load **11** via the current switching circuit **61**.

Further, the supply line **41** comprises a parallel part **41**A connected with the first switch line **42** and the second switch line **44** in parallel. The parallel part **41**A is provided on a side closer to the power source **100** than the current switching circuit **61** (that is, upstream of the current switching circuit **61**).

<First Switch Line>

The first switch line **42** connects the first switch element **21** in parallel to the parallel part **41**A of the supply line **41**. In the first switch line **42**, the first switch element **21** and the third FET **53** are provided in this order from a side closer to the power source **100**.

<First Input Line>

The first input line **43** allows input of a current from the first switch line **42** to the control circuit **31**. The first input line **43** has one end connected to the first switch line **42** between the first switch element **21** and the third FET **53**. When the first switch element **21** is closed, the current is input from the first input line **43** to the control circuit **31**.

<Second Switch Line>

The second switch line **44** connects the second switch element **22** in parallel to the parallel part **41**A of the supply line **41**. In the second switch line **44**, the second switch element **22** and the fourth FET **54** are provided in this order from a side closer to the power source **100**.

<Second Input Line>

The second input line **45** allows input of a current from the second switch line **44** to the control circuit **31**. The second input line **45** has one end connected to the second switch line **44** between the second switch element **22** and the fourth FET **54**. When the second switch element **22** is closed, the current is input from the second input line **45** to the control circuit **31**.

<FET>

The first FET **51** and the second FET **52** are general-purpose field effect transistors provided in the parallel part **41**A of the supply line **41**. The first FET **51** and the second FET **52** are connected in parallel to both of the first switch element **21** and the second switch element **22**.

The first FET **51** is switched on and off in accordance with a current input from the control circuit **31**. The first FET **51** includes a first body diode **51**A having a forward direction coincident with a direction in which a current is supplied from the power source **100** to the load **11**. Herein, the body diode is a diode arranged at a back gate.

The second FET **52**, along with the first FET **51**, is switched on and off in accordance with the current input from the control circuit **31**. The second FET **52** includes a second body diode **52**A having a forward direction opposite to the forward direction of the first body diode **51**A (that is, a reverse direction toward the power source **100**).

The first FET **51** and the second FET **52** are concurrently switched on or off by the control circuit **31**. In other words, the control circuit **31** switches between a state, in which both the first FET **51** and the second FET **52** are switched off, and a state, in which both the first FET **51** and the second FET **52** are switched on.

In the state in which both the first FET **51** and the second FET **52** are switched off, a current directed to the load **11** flows through the first FET **51**, but a current directed to the power source **100** in the reverse direction does not flow through the first FET **51**. Simultaneously, a current directed to the load **11** does not flow through the second FET **52**. As a result, no current flows through the parallel part **41**A.

In the state in which both the first FET **51** and the second FET **52** are switched on, the current flows through both of the first FET **51** and the second FET **52** toward the load **11**. As a result, the current flows through the parallel part **41**A.

The third FET **53** is a general-purpose field effect transistor provided in the first switch line **42** in parallel to the first input line **43**. Specifically, the third FET **53** is provided in the first switch line **42** on a side closer to the load **11** than a connection point P**2** between the first switch element **21** and the first input line **43**.

The third FET **53** includes a third body diode **53**A having a forward direction coincident with a direction in which a current is supplied to the load **11** via the first switch element **21**. The third FET **53** is switched on or off by the control circuit **31**. In a state in which the third FET **53** is switched off, a current directed from the first switch element **21** toward the power source **100** in the first switch line **42** in the reverse direction does not flow.

The fourth FET **54** is a general-purpose field effect transistor provided in the second switch line **44** in parallel to the second input line **45**. Specifically, the fourth FET **54** is provided in the second switch line **44** on a side closer to the load **11** than a connection point P**3** between the second switch element **22** and the second input line **45**.

The fourth FET **54** includes a fourth body diode **54**A having a forward direction coincident with a direction in which a current is supplied to the load **11** via the second switch element **22**. The fourth FET **54** is switched on or off by the control circuit **31**. In a state in which the fourth FET **54** is switched off, a current directed from the second switch element **22** toward the power source **100** in the second switch line **44** in the reverse direction does not flow.

<Current Switching Circuit>

The current switching circuit **61** is provided in the supply line **41** between a connection point P**1**, at which the first switch line **42** and the second switch line **44** are connected, and the load **11**.

The current switching circuit **61** comprises a first line **61**A, a second line **61**B, a third line **61**C, a fourth line **61**D, a fifth line **61**E, a first switching FET **61**F, a second switching FET **61**G, a third switching FET **61**H, and a fourth switching FET **61**I.

In the closed state of the first switch element **21**, the first line **61**A allows supply of a current to the second line **61**B provided with the load **11** in the first direction. The first switching FET **61**F is provided in the first line **61**A.

The third line **61**C is connected in series with the second line **61**B and the fourth line **61**D, and connects the second line **61**B to the ground. The second switching FET **61**G is provided in the third line **61**C.

In the closed state of the second switch element **22**, the fourth line **61**D allows supply of a current to the second line **61**B provided with the load **11** in the second direction. The fourth line **61**D is connected in parallel to the first line **61**A. The third switching FET **61**H is provided in the fourth line **61**D,

The fifth line **61**E is connected in series with the first line **61**A and the second line **61**B, and connects the second line **61**B to the ground. The fifth line **61**E is connected in parallel to the third line **61**C. The fourth switching FET **61**I is provided in the fifth line **61**E.

In the state in which the second switch element **22** is open, the control circuit **31** switches on the first switching FET **61**F and the second switching FET **61**G, and switches off the third switching FET **61**H and the fourth switching FET **61**I. Accordingly, the current flows through the first line **61**A, the second line **61**B, and the third line **61**C in this order.

In the state in which the second switch element **22** is closed, the control circuit **31** switches off the first switching FET **61**F and the second switching FET **61**G, and switches on the third switching FET **61**H and the fourth switching FET **61**I. Accordingly, the current flows through the fourth line **61**D, the second line **61**B, and the fifth line **61**E in this order.

<Control by Control Circuit>

As shown in FIG. **1**, the control circuit **31** switches on the first FET **51** and the second FET **52** and switches off the third FET **53** and the fourth FET **54** when both the first switch element **21** and the second switch element **22** are open, or when the control circuit **31** receives no current input from either the first input line **43** or the second input line **45**.

In this state, the current supplied from the power source **100** passes through the parallel part **41**A where the first FET **51** and the second FET **52** are provided, and flows to the load **11**. At this time, the third FET **53** and the fourth FET **54** inhibit a current from flowing from the supply line **41** into the first switch line **42** and the second switch line **44** (see broken lines of FIG. **1**). Thus, it is possible to inhibit occurrence of a short circuit in the circuit, and thus inhibit damages to the elements.

As shown in FIG. **3**, the control circuit **31** switches off the first FET **51** and the second FET **52**, switches on the third FET **53**, and switches off the fourth FET **54** when the first switch element **21** is closed, or when the control circuit **31** receives a current input from the first input line **43**. Also, the control circuit **31** controls the current switching circuit **61** to cause the current to flow through the load **11** in the first direction when the control circuit **31** receives the current input from the first input line **43**.

In this state, the current supplied from the power source **100** passes through the first switch line **42** provided with the first switch element **21**, and flows to the load **11**. At this time, the fourth FET **54** inhibits a current from flowing from the first switch line **42** into the second switch line **44** (see a broken line of FIG. **3**). Further, the first FET **51** inhibits a reverse flow of a current in the parallel part **41**A.

If the second FET **52** is not present, when the first switch element **21** in the closed state is opened, a current flows from the parallel part **41**A into the first switch element **21** until the third FET **53** switches to the off state as shown by a broken line in FIG. **4**.

In contrast, since the second FET **52** is provided in the parallel part **41**A, no current flows through the parallel part **41**A until the third FET **53** turns off and the first FET **51** and the second FET **52** turn on. That is, the second FET **52** inhibits a short circuit in the circuit when the first switch element **21** is opened.

As shown in FIG. **5**, the control circuit **31** switches off the first FET **51** and the second FET **52**, switches on the fourth FET **54**, and switches off the third FET **53** when the second switch element **22** is closed, or when the control circuit **31** receives a current input from the second input line **45**. Also, the control circuit **31** controls the current switching circuit **61** to cause the current to flow through the load **11** in the second direction when the control circuit **31** receives the current input from the second input line **45**.

In this state, the current supplied from the power source **100** passes through the second switch line **44** provided with the second switch element **22**, and flows to the load **11**. At this time, the third FET **53** inhibits a current from flowing from the second switch line **44** into the first switch line **42** (see a broken line in FIG. **5**).

When the second switch element **22** in the closed state is opened, the second FET **52** that is switched off inhibits the current from flowing into the parallel part **41**A, as in the case of opening the first switch element **21**.

1-2. Effects

The embodiment as detailed above exerts following effects.

(1a) When the first switch element **21** is closed, the first FET **51** and the second FET **52** turn off, so that the current does not flow through the parallel part **41**A but is supplied to the load **11** from the first switch line **42** provided with the first switch element **21**. This leads to interruption of the current from the power source **100** in the parallel part **41**A, so that a large current can be caused to flow through the first switch element **21**. Further, since the current is inhibited from flowing from the parallel part **41**A into the first input line **43**, the control circuit **31** can detect opening and closing of the first switch element **21**.

(1b) When the first switch element **21** is open, the third FET **53** turns off, so that the current does not flow through the first switch line **42** but is supplied from the parallel part **41**A to the load **11**. Thus, a current is inhibited from flowing from the parallel part **41**A into the first input line **43**, and the control circuit **31** can detect opening and closing of the first switch element **21**.

(1c) Since the corresponding FET is provided in each of the first switch line **42** and the second switch line **44**, it is possible to cause a large current to flow through each of the first switch element **21** and the second switch element **22**. Further, the control circuit **31** can detect opening and closing of each of the first switch element **21** and the second switch element **22**.

(1d) A selection operation between the first switch element **21** and the second switch element **22** enables a change in the direction of the current to be supplied to the load **11**.

(1e) It is possible to supply the current to the load **11** that transforms the vehicle seat **200** while removing insulating films of the first switch element **21** and the second switch element **22**.

2. Other Embodiments

Although the present embodiment has been described above, the present disclosure is not limited to the above-described embodiment and can be implemented in various modified forms.

(2a) The switch circuit in the above-described embodiment does not have to necessarily comprise the second switch element. In other words, the switch circuit may comprise only one switch element.

(2b) The switch circuit in the above-described embodiment can be used for articles other than the seat.

(2c) Functions of a single constituent element of the above-described embodiment may be distributed as a plurality of constituent elements, or functions of a plurality of constituent elements may be integrated into a single constituent element. A part of the configuration of the embodiment may be omitted. At least a part of the configuration of the embodiment may be added to or replace with another configuration of the above embodiment. Any mode included in the technical ideas specified by the language of the claims is an embodiment of the present disclosure.

What is claimed is:

1. A switch circuit, comprising:

a first switch element;

a control circuit;

a supply line configured to allow supply of a current from a power source to a load;

a first switch line that connects the first switch element in parallel to a parallel part, which is a part of the supply line;

a first input line configured to allow input of a current from the first switch line to the control circuit;

a first FET and a second FET provided in the parallel part; and

a third FET provided in the first switch line in parallel to the first input line,

wherein the first FET comprises a first body diode having a forward direction coincident with a direction in which a current is supplied to the load,

wherein the second FET comprises a second body diode having a forward direction opposite to the forward direction of the first body diode,

wherein the third FET comprises a third body diode having a forward direction coincident with a direction in which a current is supplied to the load via the first switch element, and

wherein the control circuit is configured to switch on the first FET and the second FET and switch off the third FET when the control circuit receives no current input from the first input line, and to switch off the first FET and the second FET and switch on the third FET when the control circuit receives a current input from the first input line.

2. The switch circuit according to claim **1**, further comprising:

a second switch element;

a second switch line that connects the second switch element in parallel to the parallel part;

a second input line configured to allow input of a current from the second switch line to the control circuit; and

a fourth FET provided in the second switch line in parallel to the second input line,

wherein the fourth FET comprises a fourth body diode having a forward direction coincident with a direction in which a current is supplied to the load via the second switch element, and

wherein the control circuit is configured to switch on the first FET and the second FET and switch off the third FET and the fourth FET when the control circuit receives no current input from either the first input line or the second input line, and to switch off the first FET, the second FET and the third FET and switch on the fourth FET when the control circuit receives a current input from the second input line.

3. The switch circuit according to claim **2**, further comprising a current switching circuit provided in the supply line between a connection point, at which the first switch line and the second switch line are connected, and the load,

wherein the second switch element is configured not to be closed concurrently with the first switch element, and

wherein the control circuit is configured to control the current switching circuit to cause a current to flow through the load in a first direction when the control circuit receives the current input from the first input line, and to control the current switching circuit to cause a current to flow through the load in a second direction opposite to the first direction when the control circuit receives the current input from the second input line.

4. The switch circuit according to claim **1**, wherein the switch circuit is arranged in a vehicle seat.

5. The switch circuit according to claim **2**, wherein the switch circuit is arranged in a vehicle seat.

6. The switch circuit according to claim **3**, wherein the switch circuit is arranged in a vehicle seat.

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